

CMSC 178IP: Digital Image Processing

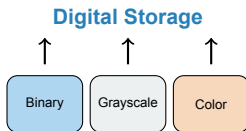
Image Types, Storage Formats, Coding & Compression

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CMSC 173 Machine Learning

Understanding Digital Image Fundamentals



Today's Learning Objectives

Core Concepts:

- ✓ **Image Types:** Binary, grayscale, color, indexed
- ✓ **Bit Depth:** 1-bit, 8-bit, 16-bit, 24-bit systems
- ✓ **Color Models:** RGB, CMYK, HSV representations
- ✓ **Resolution:** Spatial vs. intensity resolution

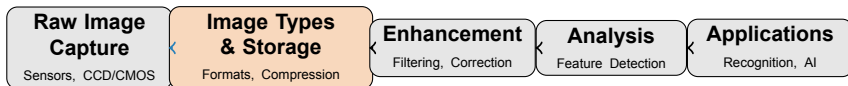
Storage & Compression:

- ✓ **Formats:** BMP, JPEG, PNG, GIF, TIFF
- ✓ **Compression:** Lossless vs. lossy techniques
- ✓ **Algorithms:** Huffman, LZW, DCT, wavelets
- ✓ **Trade-offs:** Quality vs. file size analysis

From raw pixels to optimized storage solutions!

Real-world Applications: Web graphics, medical imaging, satellite imagery, mobile photography

Digital Image Processing Pipeline



Key Questions We'll Answer Today:

- How are different image types represented in memory?
- What are the advantages/disadvantages of each storage format?
- How do compression algorithms reduce file sizes?
- When should we use lossless vs. lossy compression?

Foundation knowledge for all subsequent image processing techniques

What is a Digital Image?

Comprehensive Definition

A digital image is a **2D array of pixels** (picture elements), where each pixel contains discrete intensity values representing color or brightness information at specific spatial locations.

Mathematical Foundation:

- Image function: $f(x, y)$
- Spatial coordinates: $x \in [0, M - 1], y \in [0, N - 1]$
- Intensity levels: $L = 2^k$ (k = bit depth)
- Pixel values: $0 \leq f(x, y) \leq L - 1$

Digital Characteristics:

- **Discrete:** Finite pixel locations
- **Quantized:** Limited intensity levels
- **Rectangular:** Grid-based sampling
- **Finite:** Bounded image dimensions

Common Bit Depths & Ranges:

- 1-bit: Binary (0-1), file size = $M \times N$ bits
- 8-bit: Grayscale (0-255), file size = $M \times N$ bytes
- 24-bit: RGB Color (0-16.7M colors), file size = $3 \times M \times N$ bytes

$$\text{File Size} = M \times N \times \text{bit depth} \div 8 \text{ bytes}$$

Four Main Image Categories

1. Binary Images (1-bit)

- Values: 0 (black), 1 (white)
- Applications: OCR, barcode scanning
- Size: Smallest storage requirement
- Processing: Fast logical operations

2. Grayscale Images (8-16 bit)

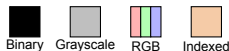
- 8-bit: 256 gray levels (0-255)
- 16-bit: 65,536 levels (medical imaging)
- Applications: X-rays, scientific imaging
- Size: 1/3 of equivalent color image

3. Color Images (24-32 bit)

- RGB: 8 bits per channel (16.7M colors)
- RGBA: Additional alpha channel
- Applications: Photography, web graphics
- Size: Largest uncompressed storage

4. Indexed Color (8-bit)

- Palette-based: 256 colors max
- Color lookup table (CLUT)
- Applications: GIF animations, icons
- Size: Reduced color palette efficiency



Selection depends on application requirements: quality, storage, processing speed

Definition & Characteristics

Binary images contain only **two intensity values**: 0 (black) and 1 (white). Each pixel requires exactly **1 bit** of storage.

Technical Specifications:

- Pixel values: $f(x, y) \in \{0, 1\}$
- Storage: $\frac{M \times N}{8}$ bytes
- Memory efficiency: 8:1 vs grayscale
- Bit packing: 8 pixels per byte

Creation Methods:

- **Thresholding:** $f(x, y) = \begin{cases} 1 & \text{if } g(x, y) > T \\ 0 & \text{otherwise} \end{cases}$
- **Dithering:** Floyd-Steinberg algorithm
- **Edge detection:** Canny, Sobel operators

Applications: OCR systems, barcode reading, medical bone analysis, industrial inspection

Advantages: Fast processing, minimal storage | Disadvantages: Loss of detail

IMAGE PLACEHOLDER

Original grayscale image

IMAGE PLACEHOLDER

Binary version (T=128)

Definition & Representation

Grayscale images represent intensity using shades of gray, typically **8-bit** (256 levels) or **16-bit** (65,536 levels).

Bit Depth Comparison:

- **8-bit:** 0-255, standard photography
- **12-bit:** 0-4095, digital cameras
- **16-bit:** 0-65535, scientific imaging
- **32-bit:** Floating point, HDR

RGB Conversion:

- **Luminance:** $Y = 0.299R + 0.587G + 0.114B$
- **Average:** $Gray = \frac{R+G+B}{3}$

Applications: Medical X-rays, satellite imagery, security systems

IMAGE PLACEHOLDER

8-bit grayscale
(256 gray levels)

IMAGE PLACEHOLDER

16-bit medical image
(65,536 levels)



Intensity Gradient

Storage: $M \times N$ bytes (8-bit) | Critical for scientific applications

RGB Fundamentals

RGB uses **additive color mixing** with Red, Green, Blue channels.
Standard: **24-bit** color (8 bits per channel).

Technical Specifications:

- Each channel: 0-255 (8-bit)
- Total colors: $256^3 = 16,777,216$
- Storage: $3 \times M \times N$ bytes
- Memory layout: Interleaved/planar

Extended Formats:

- **RGB565**: Mobile displays (16-bit)
- **RGB888**: Standard 24-bit
- **ARGB8888**: 32-bit with alpha
- **RGB48**: Professional (16-bit/channel)

Gamma Correction: $V_{out} = V_{in}^{\gamma}$ where $\gamma \approx 2.2$

Advantage: Direct display compatibility | Challenge: Large file sizes

IMAGE PLACEHOLDER

RGB Color Cube
3D Color Space

Channel Separation

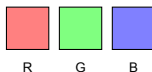


IMAGE PLACEHOLDER

Combined RGB Image

CMYK (Subtractive Model):

- Channels: Cyan, Magenta, Yellow, Black
- Application: Printing industry
- Color mixing: Ink absorption
- Smaller gamut than RGB

HSV/HSB (Perceptual Model):

- Hue: Color wheel (0-360°)
- Saturation: Purity (0-100%)
- Value: Brightness (0-100%)
- Intuitive color selection

Specialized Spaces:

- **YUV**: Video compression (luminance/chrominance)
- **LAB**: Perceptually uniform
- **XYZ**: CIE standard foundation

RGB to HSV: $V = \max(R, G, B)$, $S = \frac{V - \min(R, G, B)}{V}$

IMAGE PLACEHOLDER

RGB Color Wheel

IMAGE PLACEHOLDER

CMYK Color Model

IMAGE PLACEHOLDER

HSV Cone/Cylinder

Model selection: display (RGB), print (CMYK), perception (HSV), compression (YUV)

Concept & Implementation

Indexed color uses a **Color Lookup Table (CLUT)** with limited colors. Each pixel stores an **index** (8-bit) to the palette.

Technical Structure:

- Pixel values: 0-255 (indices)
- Palette: Up to 256 colors
- Storage: $M \times N + 256 \times 3$ bytes
- Effective when colors < 256

Palette Algorithms:

- **Median Cut:** Recursive division
- **Octree:** Tree-based quantization
- **K-means:** Color space clustering

Applications: GIF animations, retro games, scientific visualization, embedded systems

Trade-offs: Reduced file size vs limited color range, fast palette swaps vs posterization effects

Modern usage: Specialized applications with acceptable color limitations

IMAGE PLACEHOLDER

Original 24-bit Image

IMAGE PLACEHOLDER

256-color Indexed



Color Palette (256 entries)

What are Image Storage Formats?

Image storage formats define how pixel data is organized, encoded, and stored in files. They determine **compression**, **metadata handling**, and **compatibility**.

Format Classification:

- **Raster vs Vector:** Pixel-based vs mathematical shapes
- **Compressed vs Uncompressed:** File size optimization
- **Lossless vs Lossy:** Quality preservation approach
- **Single vs Multi-frame:** Static images vs animations

Key Selection Criteria:

- **File Size:** Storage and bandwidth requirements
- **Quality:** Acceptable degradation level
- **Compatibility:** Software and device support
- **Features:** Transparency, animation, metadata
- **Processing Speed:** Encode/decode performance

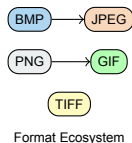
Common Format Categories:

- **Uncompressed:** BMP, PPM - Maximum quality, large files
- **Lossless Compressed:** PNG, GIF, TIFF - Quality preserved, moderate compression
- **Lossy Compressed:** JPEG, WebP - High compression, quality trade-offs
- **Professional:** RAW, TIFF, PSD - Maximum flexibility and quality

No single format is optimal for all applications - context determines the best choice

IMAGE PLACEHOLDER

Format Comparison Chart
(Size vs Quality)



BMP Characteristics

Windows Bitmap is Microsoft's native raster format, typically storing **uncompressed** pixel data with simple file structure.

Technical Specifications:

- **File Structure:** Header + Pixel data
- **Color Support:** 1, 4, 8, 16, 24, 32-bit
- **Compression:** Usually none (RLE available)
- **Max Dimensions:** Limited by 32-bit integers
- **Byte Order:** Little-endian (Intel format)

File Structure Details:

- **File Header:** 14 bytes (signature, file size)
- **Info Header:** 40+ bytes (dimensions, bit depth)
- **Color Palette:** Optional for ≤ 8 -bit images
- **Pixel Data:** Row-by-row, bottom-to-top
- **Row Padding:** Aligned to 4-byte boundaries

Advantages: Simple structure, no compression artifacts, universal Windows support, fast loading

Disadvantages: Very large file sizes, limited web support, no transparency support, minimal metadata

Use Cases: System icons, wallpapers, simple graphics, temporary image storage, legacy applications

IMAGE PLACEHOLDER

BMP File Structure
Diagram

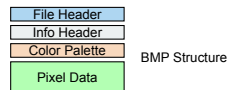


IMAGE PLACEHOLDER

Size Comparison
BMP vs Other Formats

JPEG Fundamentals

Joint Photographic Experts Group format uses **DCT-based lossy compression** optimized for photographic images with smooth color transitions.

Compression Pipeline:

- **Color Space:** RGB $\rightarrow$ YCbCr conversion
- **Subsampling:** Reduce chrominance resolution
- **DCT Transform:** 8 $\times$ 8 block frequency analysis
- **Quantization:** Discard high-frequency data
- **Huffman Coding:** Entropy-based compression

Quality Control:

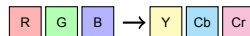
- **Quality Setting:** 1-100 scale (higher = better)
- **Quantization Tables:** Control frequency removal
- **Compression Ratio:** 10:1 to 50:1 typical
- **Artifacts:** Blocking, ringing, mosquito noise

Technical Features:

- **Progressive:** Interlaced loading capability
- **EXIF Data:** Camera metadata support
- **Color Profiles:** ICC profile embedding

IMAGE PLACEHOLDER

JPEG Quality Comparison
(100% vs 50% vs 10%)



RGB $\rightarrow$ YCbCr

IMAGE PLACEHOLDER

DCT 8 $\times$ 8 Blocks
Visualization

PNG Overview

Portable Network Graphics provides **lossless compression** with advanced features like transparency and gamma correction, designed as a GIF replacement.

Technical Specifications:

- **Color Types:** Grayscale, RGB, Palette, Grayscale+Alpha, RGB+Alpha
- **Bit Depths:** 1, 2, 4, 8, 16 bits per channel
- **Compression:** DEFLATE algorithm (LZ77 + Huffman)
- **Interlacing:** Adam7 progressive display
- **Transparency:** Full alpha channel support

PNG Chunk Structure:

- **Critical Chunks:** IHDR, PLTE, IDAT, IEND
- **Ancillary Chunks:** tEXt, gAMA, cHRM, tIME
- **CRC Protection:** Error detection for each chunk
- **Extensible:** Custom chunk support

Compression Techniques:

- **Filtering:** Pre-processing for better compression
- **DEFLATE:** Dictionary-based compression
- **Palette Optimization:** Reduce colors when possible

IMAGE PLACEHOLDER

PNG Transparency Demo
Alpha Channel Example

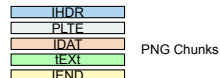


IMAGE PLACEHOLDER

Compression Comparison
PNG vs JPEG vs BMP

GIF - Graphics Interchange Format

Legacy format with 256-color limitation, famous for **animation support** and **LZW compression**.

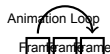
GIF Characteristics:

- **Color Limit:** 256 colors per frame
- **Compression:** LZW lossless algorithm
- **Animation:** Multiple frame support
- **Transparency:** 1-bit (on/off only)
- **Interlacing:** Progressive display

GIF Examples:

IMAGE PLACEHOLDER

GIF Animation Frames



TIFF Examples:

IMAGE PLACEHOLDER

TIFF Multi-page Document

IMAGE PLACEHOLDER

Professional Photography
TIFF Example

TIFF - Tagged Image File Format

Professional format supporting multiple compression methods, extensive metadata, and multi-page documents.

TIFF Features:

- **Flexibility:** Multiple compression options
- **Color Support:** 1-bit to 32-bit per channel
- **Metadata:** Extensive tag system
- **Multi-page:** Document scanning support

What is Image Coding?

Image coding is the process of converting continuous visual information into **discrete digital data** that can be stored, transmitted, and processed by computers.

Coding Process Steps:

- **Sampling:** Convert continuous spatial domain to discrete pixels
- **Quantization:** Map continuous intensity to discrete levels
- **Encoding:** Represent pixel values in binary format
- **Storage/Transmission:** Organize data for file systems

Key Coding Parameters:

- **Spatial Resolution:** Pixels per inch (PPI/DPI)
- **Intensity Resolution:** Bits per pixel (bit depth)
- **Color Resolution:** Number of color channels
- **Temporal Resolution:** Frames per second (video)

Coding Challenges:

- **Storage Requirements:** Uncompressed files are massive
- **Transmission Bandwidth:** Network limitations
- **Processing Speed:** Real-time requirements
- **Quality vs Size:** Trade-off optimization

IMAGE PLACEHOLDER

Analog to Digital
Conversion Process



Sampled & Quantized

Continuous Signal

IMAGE PLACEHOLDER

Bit Depth Comparison
(1, 8, 16, 24-bit)

Color Model Selection Impact

Different color models affect **storage efficiency**, **processing complexity**, and **compression effectiveness** in image coding systems.

RGB Color Coding:

- **Standard:** 8 bits per channel (24-bit total)
- **Range:** Each channel 0-255
- **Storage:** 3 bytes per pixel
- **Advantage:** Direct display mapping
- **Disadvantage:** Correlated channels

YUV/YCbCr for Compression:

- **Y:** Luminance (brightness) - 8 bits
- **U/Cb:** Blue chrominance - 8 bits
- **V/Cr:** Red chrominance - 8 bits
- **Advantage:** Separates luminance/color
- **Compression:** Chrominance subsampling

Chroma Subsampling Schemes:

- **4:4:4** - No subsampling (full quality)
- **4:2:2** - Horizontal chrominance reduction
- **4:2:0** - Both horizontal and vertical reduction

IMAGE PLACEHOLDER

RGB vs YUV
Channel Comparison

 4:4:4
 4:2:2
 4:2:0

IMAGE PLACEHOLDER

Subsampling Effects
Visual Quality Impact

Compression Necessity

Image compression reduces file sizes by **eliminating redundancy** and **exploiting perceptual limitations** of human vision.

Types of Redundancy:

- **Spatial:** Neighboring pixels often similar
- **Spectral:** Color channels correlated
- **Temporal:** Video frame similarities
- **Psychovisual:** Human eye limitations

Compression Classification:

- **Lossless:** Perfect reconstruction possible
- **Lossy:** Some information permanently lost
- **Near-lossless:** Small controlled loss
- **Visually lossless:** Imperceptible quality loss

Performance Metrics:

- **Compression Ratio:** Original size / Compressed size
- **Bit Rate:** Bits per pixel (bpp)
- **PSNR:** Peak Signal-to-Noise Ratio
- **SSIM:** Structural Similarity Index

Compression Techniques Overview:

IMAGE PLACEHOLDER

Compression Ratio
vs Quality Graph

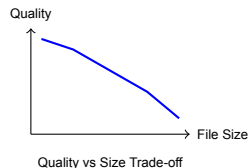


IMAGE PLACEHOLDER

Lossless vs Lossy
Comparison Example

Lossless Compression Principles

Lossless compression achieves **perfect reconstruction** by exploiting statistical redundancy without discarding information.

Huffman Coding:

- **Principle:** Variable-length codes based on frequency
- **Algorithm:** Build binary tree from symbol probabilities
- **Efficiency:** Optimal for symbol-by-symbol coding
- **Applications:** JPEG (AC coefficients), PNG
- **Limitation:** Requires probability knowledge

LZW (Lempel-Ziv-Welch):

- **Principle:** Dictionary-based pattern matching
- **Algorithm:** Build dictionary during encoding
- **Advantage:** Adaptive, no prior knowledge needed
- **Applications:** GIF, TIFF, PostScript
- **Performance:** Good for images with patterns

Run-Length Encoding (RLE):

- **Principle:** Replace repeated values with count
- **Example:** AAABBBBCC → 3A4B2C
- **Efficiency:** Excellent for binary images

IMAGE PLACEHOLDER

Huffman Tree
Construction Example

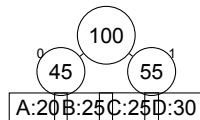


IMAGE PLACEHOLDER

LZW Dictionary
Building Process

IMAGE PLACEHOLDER

RLE Example
Binary Image



Lossy Compression Philosophy

Lossy compression achieves **high compression ratios** by permanently removing information that has **minimal perceptual impact**.

Transform-Based Compression:

- **DCT (Discrete Cosine Transform):** JPEG standard
- **Principle:** Convert spatial to frequency domain
- **8×8 Blocks:** Process image in small segments
- **Quantization:** Remove high-frequency components
- **Entropy Coding:** Huffman encode remaining data

Wavelet-Based Compression:

- **JPEG 2000:** Modern wavelet standard
- **Multi-resolution:** Analyze multiple scales
- **Better Quality:** Less blocking artifacts
- **Progressive:** Quality/resolution scalability
- **Applications:** Medical, satellite imagery

Quantization Strategies:

- **Uniform:** Equal step sizes
- **Non-uniform:** Perceptually-based steps
- **Adaptive:** Content-dependent adjustment

IMAGE PLACEHOLDER

DCT Frequency
Coefficients 8×8



8×8 DCT Block

IMAGE PLACEHOLDER

Wavelet Decomposition
Multi-scale Analysis

IMAGE PLACEHOLDER

Compression Artifacts
Blocking vs Ringing



Modern Compression Standards

Contemporary image compression algorithms balance **compression efficiency**, **computational complexity**, and **visual quality** for specific applications.

Next-Generation Formats:

- **WebP:** Google's VP8-based format
- **HEIF:** High Efficiency (H.265-based)
- **AVIF:** AV1-based open standard
- **JPEG XL:** Latest JPEG committee standard

Performance Metrics Comparison:

- **Compression Ratio:** AVIF > HEIF > WebP > JPEG
- **Quality (SSIM):** JPEG XL > AVIF > HEIF > WebP
- **Encode Speed:** JPEG > WebP > HEIF > AVIF
- **Decode Speed:** JPEG > WebP > HEIF > AVIF
- **Browser Support:** JPEG > WebP > AVIF > HEIF

Specialized Applications:

- **Medical Imaging:** JPEG-LS, JPEG 2000 (lossless)
- **Satellite Data:** Custom wavelets, CCSDS standards
- **Mobile Photography:** HEIF with computational photography

IMAGE PLACEHOLDER

Compression Efficiency
Comparison Chart

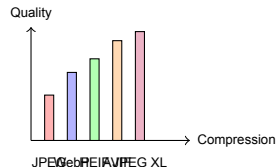


IMAGE PLACEHOLDER

Visual Quality
at Same File Size

IMAGE PLACEHOLDER

Medical Imaging Systems:

- **DICOM Standard:** JPEG-LS, JPEG 2000 lossless
- **Requirements:** No diagnostic information loss
- **Storage:** Petabytes of patient data
- **Transmission:** PACS networks, telemedicine
- **Example:** 16-bit CT scans, 12-bit X-rays

Social Media Platforms:

- **Facebook:** WebP for web, HEIF mobile uploads
- **Instagram:** Aggressive JPEG compression with AI enhancement
- **TikTok:** H.264 video, JPEG thumbnails
- **Challenges:** Billions of images, mobile bandwidth
- **Innovation:** Neural super-resolution post-processing

E-commerce & Product Photography:

- **Amazon:** Multi-format delivery (WebP, JPEG, AVIF)
- **Requirements:** Fast loading, zoom capability
- **Progressive Loading:** Low-quality → high-quality
- **Mobile Optimization:** Responsive images, lazy loading

Scientific & Satellite Imagery:

- **NASA:** Custom wavelets, lossless compression
- **Google Earth:** Multi-resolution pyramids

IMAGE PLACEHOLDER

Medical DICOM
Viewer Interface

IMAGE PLACEHOLDER

Social Media
Image Pipeline

IMAGE PLACEHOLDER

E-commerce
Progressive Loading

IMAGE PLACEHOLDER

Satellite Imagery

What We've Learned Today

Comprehensive understanding of digital image representation, storage formats, and compression techniques that form the foundation of modern digital imaging.

Core Concepts Mastered:

- **Image Types:** Binary, grayscale, color, indexed representations
- **Storage Formats:** BMP, JPEG, PNG, GIF, TIFF characteristics
- **Color Models:** RGB, YUV, color space conversions
- **Compression:** Lossless vs lossy trade-offs and algorithms

Critical Decision Framework:

- **Quality Requirements:** Medical = lossless, Web = acceptable loss
- **Storage Constraints:** Mobile vs server storage capabilities
- **Processing Power:** Real-time vs offline compression
- **Compatibility:** Universal support vs cutting-edge efficiency

Practical Guidelines:

- **Photography:** JPEG for sharing, RAW+lossless for archival
- **Graphics/Logos:** PNG for transparency, SVG for scalability
- **Web Development:** Progressive enhancement (AVIF→WebP→JPEG)

● **Scientific Data:** Always preserve original, compress copies

IMAGE PLACEHOLDER

Digital Imaging
Ecosystem Overview

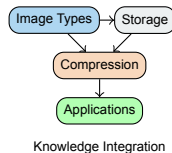


IMAGE PLACEHOLDER

Future Technology
Roadmap Timeline